

भारतीय मानक

IS 1077 : 2025

Indian Standard

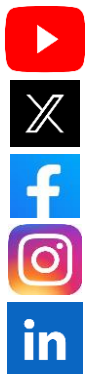
सामान्य जली हुई मिट्टी की इमारत
ईंटें — विशिष्ट
(छठा पुनरीक्षण)

Common Burnt Clay Building
Bricks — Specification
(Sixth Revision)

ICS 91.100.25

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BUREAU OF INDIAN STANDARDS
मानक भवन, 9 बहादुर शाह ज़फर मार्ग, नई दिल्ली - 110002
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI - 110002
www.bis.gov.in www.standardsbis.in

August 2025

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Clay and Stabilized Soil Products for Construction Sectional Committee, CED 30

FOREWORD

This Indian Standard (Sixth Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Clay and Stabilized Soil Products for Construction Sectional Committee had been approved by the Civil Engineering Division Council.

The common building bricks is not only one of the oldest but also the most extensively used building material in construction work. It is essentially a local building material and consequently there exist considerable variations in the quality of raw material. The manufacturing process and the quality of the finished product. Rapid building activity, to be on rational lines, needs a certain degree of uniformity in the construction materials. Standardization of the common building brick with regard to its quality and dimensions would substantially help in raising the quality of construction work and its speed. This standard was first published in 1957 and subsequently revised in 1966, 1970, 1976, 1986 and 1992.

Keeping in view the advantages of modular co-ordination. Indian standards specify the dimensions of standard bricks in 100 mm module as the basis of all dimensional standardization in regard to building components. This is also in conformity with the decision of Government of India to adopt metric system in the country. Considering the various issues regarding the manufacturing and other practices followed in the country, the sectional committee responsible for the preparation of this standard had specified modular size of the brick. Advantages that a modular brick has over traditional brick are many, such as:

- a) Requires less drying area;
- b) Increased floor space;
- c) Reduction in construction labour cost;
- d) Reduction in losses during handling; and
- e) Reduction in mortar consumption.

The significant modifications in this revision are:

- a) The units have been modified to SI units;
- b) The size of modular bricks has been updated;
- c) The list of non-modular sizes of bricks has been updated;
- d) The condition for warpage has been included;
- e) The requirement for modulus of rupture has been included; and
- f) The requirement for initial rate of absorption values have been included.

In the revision of this standard, due weightage has been given to international co-ordination among the standards and practices prevailing in different countries in addition to relating it to the construction practices in India.

This standard contributes to the United Nations Sustainable Development Goal 11 'Sustainable cities and communities' towards strengthen efforts to protect and safeguard the world's cultural and natural heritage and Goal 12 'Ensure sustainable consumption and production patterns' towards substantially reduce waste generation through prevention, reduction, recycling and reuse.

The composition of the Committee responsible for revision of this standard is given in [Annex A](#).

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard
**COMMON BURNT CLAY BUILDING BRICKS —
SPECIFICATION**
(*Sixth Revision*)

1 SCOPE

This standard lays down requirements for classification, general quality, dimensions and physical requirements of common burnt clay building bricks used in buildings.

NOTE — Only burnt clay bricks having compressive strength less than 40 N/mm² are covered in this standard; for higher strength bricks, see IS 2180 : 2025 'Specification for heavy duty burnt clay building bricks' (*fourth revision*).

2 REFERENCES

The standards given below contain provisions which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards.

IS No.	Title
IS 2248 : 2025	Glossary products for buildings — Glossary of terms (<i>third revision</i>)
IS 3495	Burnt clay building bricks — Methods of tests
(Part 1 : 2019)	Determination of compressive strength (<i>fourth revision</i>)
(Part 2 : 2019)	Determination of water absorption (<i>fourth revision</i>)
(Part 3 : 2019)	Determination of efflorescence (<i>fourth revision</i>)
(Part 1 : 2019)	Determination of warpage (<i>fourth revision</i>)

IS No.

Title

(Part 5) : Determination of initial rate of absorption

(Part 6) : Determination of modulus of rupture

IS 5454 : Burnt clay bricks and burnt clay tiles — Methods of sampling (*second revision*)

3 TERMINOLOGY

For the purpose of this standard, the definitions as given in IS 2248 shall apply.

4 CLASSIFICATION

The common burnt clay bricks shall be classified on the basis of average compressive strength as given in [Table 1](#).

5 GENERAL QUALITY

5.1 Bricks shall be hand-moulded or machine-moulded and shall be made from suitable soils. They shall be free from cracks and flaws and nodules of free lime.

5.2 Hand-moulded bricks of 90 mm or 70 mm height shall be moulded with a frog 10 mm to 20 mm deep on one of its flat sides; the shape and size of the frog shall conform to either [Fig. 1\(A\)](#) or [Fig. 1\(B\)](#) (see [6.1.1](#) for L, W and H). Bricks of 40 mm height as well as those made by extrusion process may not be provided with frogs.

5.3 The bricks shall have plain rectangular faces with sharp corners and shall be uniform in colour.

To access Indian Standards click on the link below:

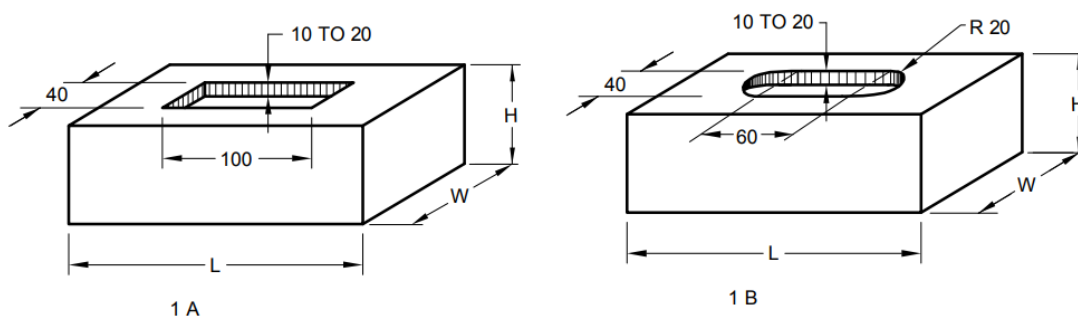
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Table 1 Classes of Common Burnt Clay Bricks

(Clause 4)

Sl No.	Class Designation	Minimum Average Compressive Strength in N/mm ²
(1)	(2)	(3)
i)	35	35.0
ii)	30	30.0
iii)	25	25.0
iv)	20	20.0
v)	17.5	17.5
vi)	15	15.0
vii)	12.5	12.5
viii)	10	10.0
ix)	7.5	7.5
x)	5	5.0
xi)	3.5	3.5



All dimensions in millimetres.

FIG. 1 SHAPE AND SIZE OF FROGS IN BRICKS

6 DIMENSIONS AND TOLERANCES

6.1 Dimensions

6.1.1 The standard modular size of common building bricks shall be as follows [Fig. 1(A) and Fig. 1(B)]:

Sl No.	Length (L)	Width (W)	Height (H)
	mm	mm	mm
(1)	(2)	(3)	(4)
i)	190	90	90
ii)	190	90	40

6.1.2 The following non-modular sizes of the bricks

may also be used [Fig. 1(A) and Fig. 1(B)]:

Sl No.	Length (L)	Width (W)	Height (H)
	mm	mm	mm
(1)	(2)	(3)	(4)
i)	230	110	75
ii)	230	110	70
iii)	230	150	70
iv)	255	120	75

6.2 Tolerances

6.2.1 The dimensions of bricks when tested in accordance with 6.2.2 shall be within the following limits per 20 bricks:

Sl No.	Length/Width/Height		For Modular Sizes	For Non-Modular Sizes	Tolerance
(1)	(2)		(3)	(4)	(5)
i)	Length, mm	in	3 800	4 600	± 80
ii)	Width, mm	in	1 300	2 200	± 40
iii)	Height, mm	in	—	—	—
iv)	For 90mm high bricks		1 800	1 400	± 40
v)	For 40 mm high bricks		800	600	± 40

6.2.2 Twenty (or more according to the size of stack) whole bricks shall be selected at random from the sample selected under 8. All blisters, loose particles of clay and small projections shall be removed. They shall then be arranged upon a level surface successively as indicated in [Fig. 2(A), Fig. 2(B) and Fig. 2(C)] in contact with each other and in a straight line. The overall length of the assembled bricks shall be measured with a steel tape or other suitable inextensible measure sufficiently long to measure the whole row at one stretch. Measurement by repeated application of

short rule or measure shall not be permitted. If for any reason it is found impracticable to measure bricks in one row, the sample may be divided into rows of 10 bricks each which shall be measured separately to the nearest millimetre. All these dimensions shall be added together.

7 PHYSICAL REQUIREMENTS

7.1 Compressive Strength

The bricks, when tested in accordance with the procedure laid down in IS 3495 (Part 1) shall have a minimum average compressive strength for various classes as given in 4.

The compressive strength of any individual brick tested shall not fall below the minimum compressive

strength specified for the corresponding class of brick by more than 15 percent. The lot shall be then checked for next lower class of brick

7.2 Water Absorption

The bricks, when tested in accordance with the procedure laid down in 4.1 of IS 3495 (Part 2) after immersion in water for 24 h, water absorption shall not be more than 20 percent by dry weight up to class 12.5 and 15 percent by weight for higher classes.

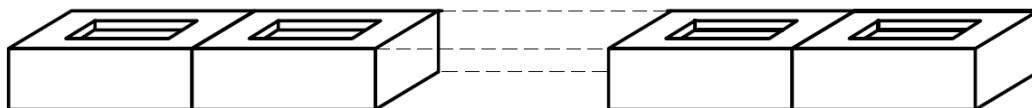


FIG 2 A MEASUREMENT OF LENGTH

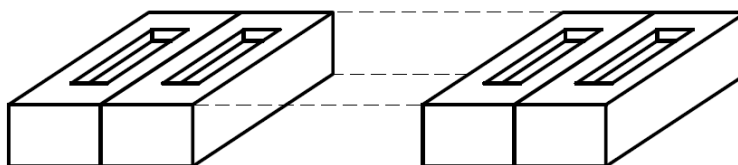


FIG 2 B MEASUREMENT OF WIDTH

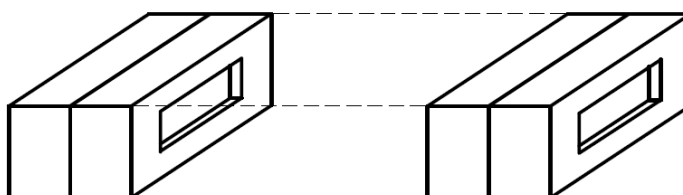


FIG 2 C MEASUREMENT OF HEIGHT

FIG. 2 MEASUREMENT OF TOLERANCES OF COMMON BUILDING BRICKS

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7.3 Efflorescence

The bricks when tested in accordance with the procedure laid down in IS 3495 (Part 3) the rating of efflorescence shall not be more than 'moderate' up to class 12.5 and 'slight' for higher classes.

7.4 Warpage (Edge curvature)

The bricks when tested in accordance with the procedure laid down in IS 3495 (Part 4) the average warpage shall not exceed 3 mm or 5 percent of the brick height, whichever is lower.

7.5 Modulus of Rupture

The bricks when tested as per the procedure laid down in IS 3495 (Part 5) the average value of the modulus of rupture to its compressive strength shall be more than 0.07.

7.6 Initial Rate of Absorption (IRA)

The bricks when tested as per the procedure laid

down in IS 3495 (Part 6) the average value of IRA shall be less than 40 N/m²/min.

8 SAMPLING AND CRITERION FOR CONFORMITY

Sampling and criterion for conformity of common bricks shall be done in accordance with the procedure laid down in IS 5454.

9 MARKING

Each brick shall be marked (in the frog if provided) with an indication of source of manufacture.

9.1.1 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

ANNEX A

(Foreword)

COMMITTEE COMPOSITION

Clay and Stabilized Soil Products for Construction Sectional Committee, CED 30

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In Personal Capacity (<i>Flat 119, Amulya Fortune Apartments, Road No 1, Opp. Union Bank, Madhavapuri Hills, Chandanagar, Hyderabad - 500050</i>)	SHRI K. SITARAMANANJEYULU
BIS Directorate General	SHRI DWAIPAYAN BHADRA, SCIENTIST 'E'/DIRECTOR AND HEAD (CIVIL ENGINEERING) [REPRESENTING DIRECTOR GENERAL (<i>Ex-officio</i>)]

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Amendments Issued Since Publication

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BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002

Telephones: 2323 0131, 2323 3375, 2323 9402

Website: www.bis.gov.in

Regional Offices:

	Telephones
Central : 601/A, Konnectus Tower -1, 6 th Floor, DMRC Building, Bhavbhuti Marg, New Delhi 110002	{ 2323 7617
Eastern : 8 th Floor, Plot No 7/7 & 7/8, CP Block, Sector V, Salt Lake, Kolkata, West Bengal 700091	{ 2367 0012 2320 9474
Northern : Plot No. 4-A, Sector 27-B, Madhya Marg, Chandigarh 160019	{ 265 9930
Southern : C.I.T. Campus, IV Cross Road, Taramani, Chennai 600113	{ 2254 1442 2254 1216
Western : 5 th Floor/MTNL CETTM, Technology Street, Hiranandani Gardens, Powai, Mumbai 400076	{ 2570 0030 2570 2715

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